INTRODUCTION

To understand why Richard Feynman was such a great teacher, it is important to appreciate his remarkable stature as a scientist. He was indeed one of the outstanding figures of twentieth-century theoretical physics. His contributions to that subject are central to the whole development of the particular way in which quantum theory is used in current cutting-edge research and thus to our present-day pictures of the world. The Feynman path integrals, Feynman diagrams, and Feynman rules are among the very basic tools of the modern theoretical physicist—tools that are necessary for the application of the rules of quantum theory to physical fields (e.g., the quantum theory of electrons, protons, and photons), and which form an essential part of the procedures whereby one makes these rules consistent with the requirements of Einstein's Special Relativity theory. Although none of these ideas is easy to appreciate, Feynman's particular approach always had a deep clarity about it, sweeping away unnecessary complications in what had gone before. There was a close link between his special ability to make progress in research and his particular qualities as a teacher. He had a unique talent that enabled him to cut through the complications that often obscure the essentials of a physical issue and to see clearly into the deep underlying physical principles.

Yet, in the popular conception of Feynman, he is known more for his antics and buffoonery, for his practical jokes, his irreverence